

UHF RFID Patch Antenna Design

Circularly Polarized Array for Vehicle Access Control (Boom Barrier Lane)

RF Engineer Internship — Technical Assignment Report

Submitted by Arya

arya26062005@gmail.com

<https://github.com/Aryaawe>

1. Task Overview

This report documents the design, simulation, and analysis of a circularly polarized UHF RFID patch antenna intended for a boom-barrier / lane-based vehicle access control system.

Frequency band	865–868 MHz (ETSI UHF RFID)
Target gain	8–9 dBi
Polarization	Circular
Application	Vehicle access control, boom barrier lane
Key constraint	Narrow beam, minimal side/adjacent-lane reads
Mounting	Metal pole

2. Design Approach

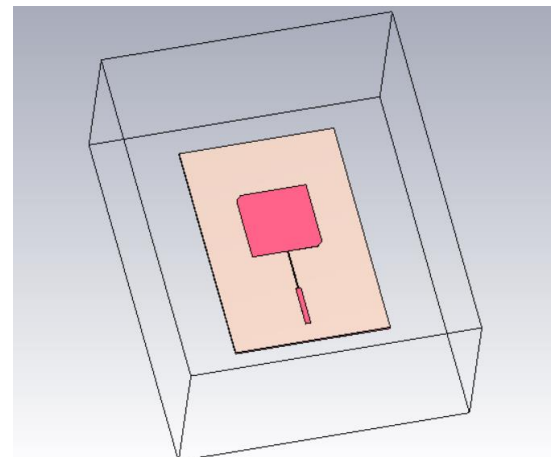
2.1 Antenna Type Selected

- **Square patch:** Low-profile, planar, and easy to mount on a pole.
- **Corner truncation:** Produces circular polarization using a single feed.
- **Quarter-wave transformer:** Provides **50 Ω impedance matching** without a coaxial probe.
- **Array factor analysis:** CST's **Farfield Array Factor** was used to study the effect of combining two identical elements, showing increased directivity and a narrower beam, which helps reduce side reads.

2.2 Substrate Selection — Rogers RO4003C

Rogers RO4003C ($\epsilon_r = 3.38$, $\tan\delta \approx 0.0027$) was used in place of FR4 for this design. Justification:

- **Low loss:** Improves antenna efficiency and gain.
- **Stable ϵ_r :** Keeps the resonant frequency stable outdoors.
- **Low moisture absorption:** Reduces frequency changes due to weather.
- **Easy fabrication:** Can be manufactured similarly to FR4.
- **Moderate ϵ_r (3.38):** Keeps the antenna size practical while maintaining bandwidth.



2.3 Wavelength Calculation

At the design center frequency $f_0 = 866.5$ MHz:

$$\lambda_0 = c / f_0 = (3 \times 10^8 \text{ m/s}) / (866.5 \times 10^6 \text{ Hz}) \approx 346.2 \text{ mm}$$

Across the band this varies only slightly: $\lambda_0 \approx 346.7$ mm at 865 MHz and ≈ 345.6 mm at 868 MHz — confirming the patch can be designed once at the center frequency without a multi-point re-optimization.

2.4 Patch Dimension Estimation ($\approx \lambda/2$)

The standard transmission-line model gives a first-order width/length estimate before corner truncation is applied:

$$W = c / (2f_0) \times \text{sqrt}[2 / (\epsilon_r + 1)]$$

$$\epsilon_{r,eff} = (\epsilon_r + 1)/2 + (\epsilon_r - 1)/2 \times [1 + 12h/W]^{-0.5}$$

$$\Delta L = 0.412h \times (\epsilon_{r,eff} + 0.3)(W/h + 0.264) / [(\epsilon_{r,eff} - 0.258)(W/h + 0.8)]$$

$$L = \lambda_0 / (2 \times \text{sqrt}[\epsilon_{r,eff}]) - 2\Delta L$$

- **Initial calculation:** For RO4003C with $h \approx 1.5$ mm, the calculated patch dimensions were approximately **W = 117 mm** and **L = 94 mm**.
- **Square patch:** For circular polarization, the patch was changed to a **square shape (W $\approx$ L)** to support two nearly identical orthogonal modes.
- **Corner truncation:** Two opposite corners were cut to create the required **phase difference between the modes**, producing circular polarization.
- **Simulation tuning:** The truncation size and patch dimensions were adjusted in CST to improve **S11 and axial ratio** at the target frequency.
- **Final dimensions:** Therefore, the final CST dimensions are slightly different from the initial theoretical calculations.

2.5 Final Simulated Dimensions

The dimensions below are the values used in the electromagnetic simulation (CST Studio Suite), following the hand-calculated starting point and simulation-based tuning of the corner truncation and feed transformer for S11, VSWR and axial-ratio performance across 865–868 MHz:

Parameter	Dimension (mm)	Definition
mt	0.035	Copper thickness
wm	1.238	Microstrip width
st	2.9	Substrate thickness
tc	6.7	Truncation/corner cut dimension
wf	6.715	Feed width
k	7.64	Feed/patch transition dimension
lf	52.95	Feed length
lm	55.71	Patch length
pw	91.5	Patch width
pl	91.5	Patch length/overall patch dimension
sw	200	Substrate width
sl	300	Substrate length

2.6 Methods Used to Increase Gain

- Low-loss RO4003C substrate instead of FR4 to reduce dielectric loss and raise radiation efficiency.
- 4-element array combining (feed network in phase) — raised directivity from 7.17 dBi (single element) to 9.78 dBi.
- Large ground plane (300 mm × 200 mm) to reduce back-lobe leakage and diffraction loss, recovering forward gain.
- Corner-truncation size kept to the minimum needed for an acceptable axial ratio, sin

2.7 Methods Used to Narrow the Beam

- **Antenna array:** Two patches create a narrower, more focused beam.
- **Large ground plane:** Reduces unwanted radiation in other directions.
- **Symmetrical layout:** Keeps the main beam pointed forward.
- **Low side lobes:** Reduces radiation reaching nearby lanes.

3. Simulation Results

All results below are from full-wave electromagnetic simulation (CST Studio Suite) of the RO4003C corner-truncated patch, fed via the microstrip quarter-wave transformer described above.

3.1 Return Loss (S11)

The structure resonates near 860 MHz with S11 below −25 dB, and across the ETSI operating band it maintains a comfortable match:

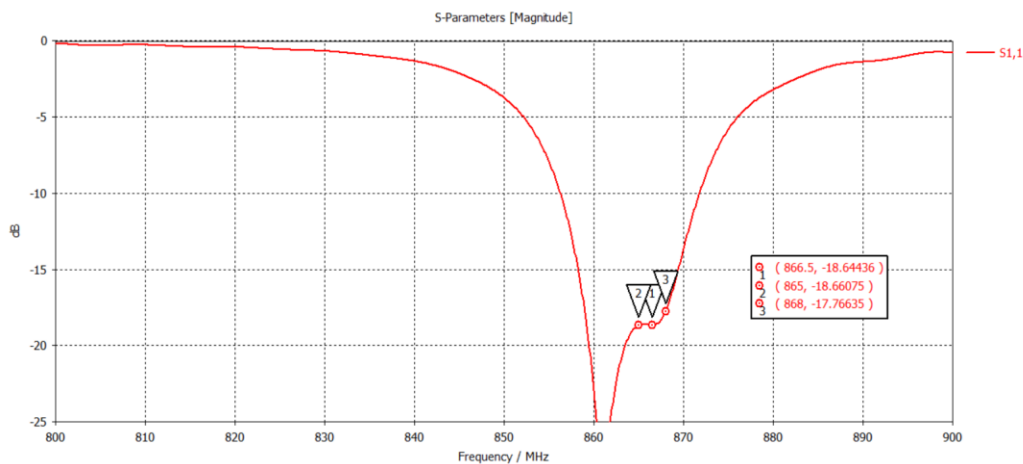


Figure 1 — S-parameter (S11) magnitude, 800–900 MHz

S11 = −18.66 dB at 865 MHz, −18.64 dB at 866.5 MHz, and −17.77 dB at 868 MHz — well below the −10 dB matching criterion across the full 865–868 MHz band.

3.2 Voltage Standing Wave Ratio (VSWR)

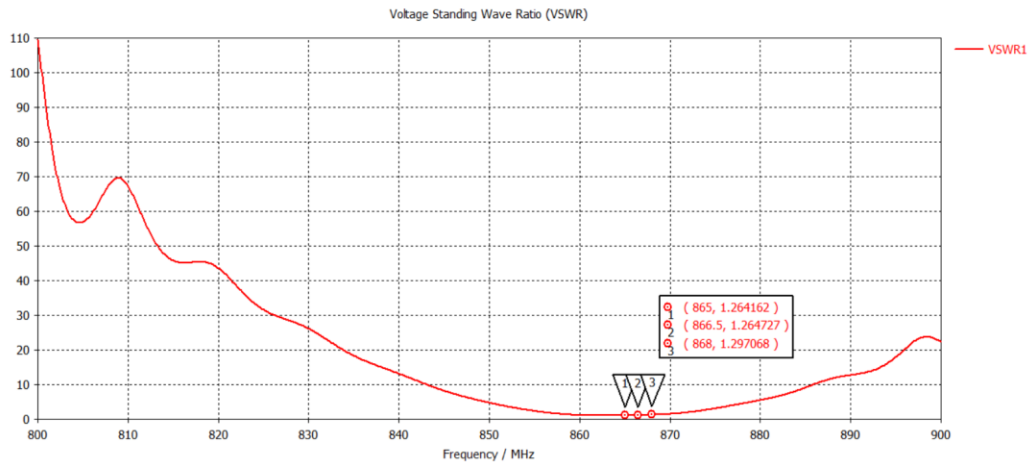


Figure 2 — VSWR, 800–900 MHz

VSWR stays between 1.26 and 1.30 across the band (1.264 at 865 MHz, 1.265 at 866.5 MHz, 1.297 at 868 MHz), well under the $\text{VSWR} < 2$ design target.

3.3 Axial Ratio (Circular Polarization Quality)

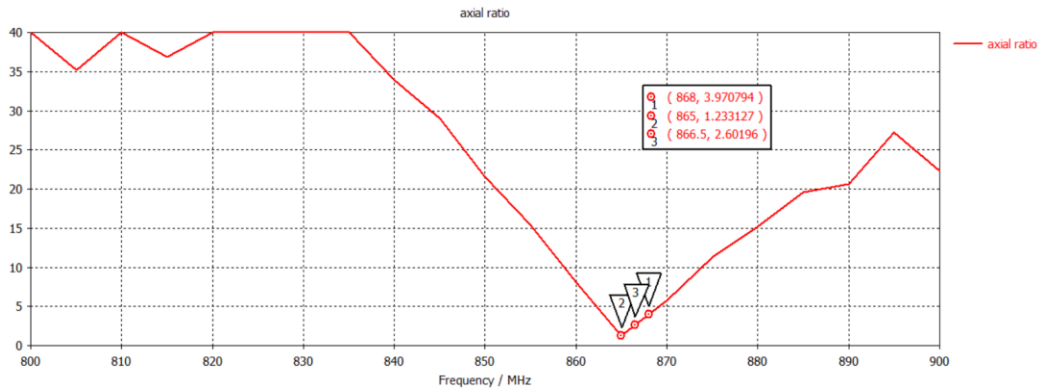


Figure 3 — Axial ratio vs. frequency, 800–900 MHz

Axial ratio is 1.23 dB at 865 MHz and 2.60 dB at 866.5 MHz, comfortably inside the 3 dB CP criterion, rising to 3.97 dB at the 868 MHz band edge. CP quality is therefore strongest at the lower half of the ETSI band and slightly outside the 3 dB criterion at the upper edge — acceptable for RFID link margin but worth a small truncation re-tune (increasing tc marginally) if uniform sub-3 dB axial ratio across the full band is required.

Parameter	865 MHz	866.5 MHz (center)	868 MHz
S11 (dB)	−18.66	−18.64	−17.77
VSWR	1.26	1.26	1.30
Axial ratio (dB)	1.23	2.60	3.97

3.4 Radiation Pattern — Single Element vs. Array

Directivity/gain summary at 866.5 MHz:

Configuration	Directivity	Radiation Efficiency	Total Efficiency
Single patch element	7.165 dBi	-1.416 dB (~72%)	-1.476 dB (~71%)
4-element array	9.780 dBi	-1.416 dB (~72%)	-1.476 dB (~71%)

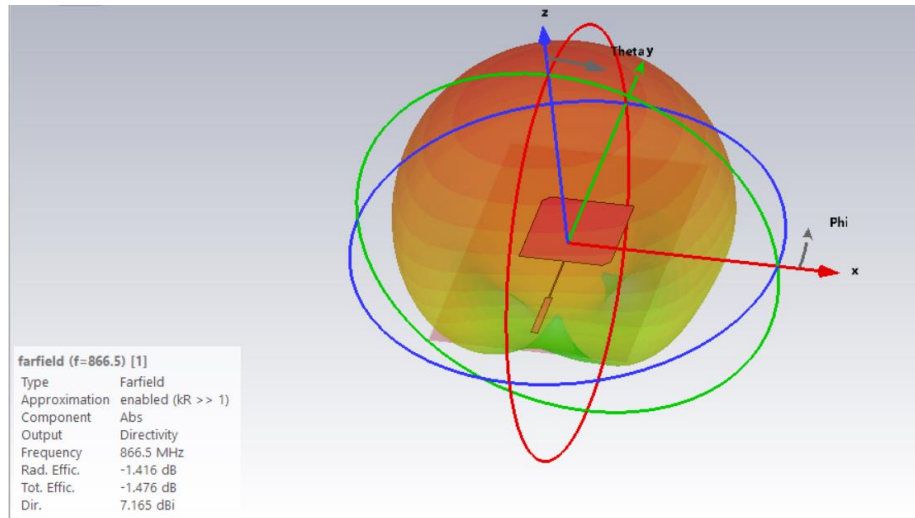


Figure 4 — 3D far-field, single patch element: 7.165 dBi

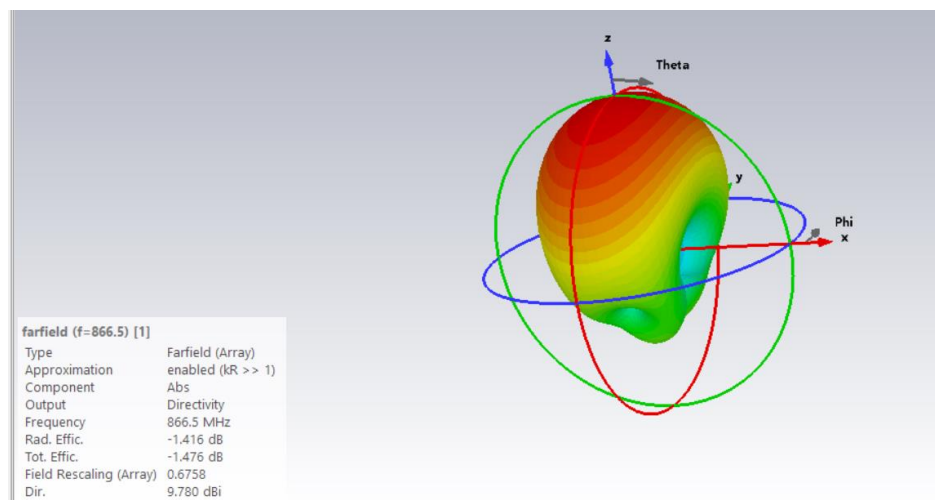


Figure 6 — 3D far-field, 4-element array (rescale 0.676): 9.780 dBi

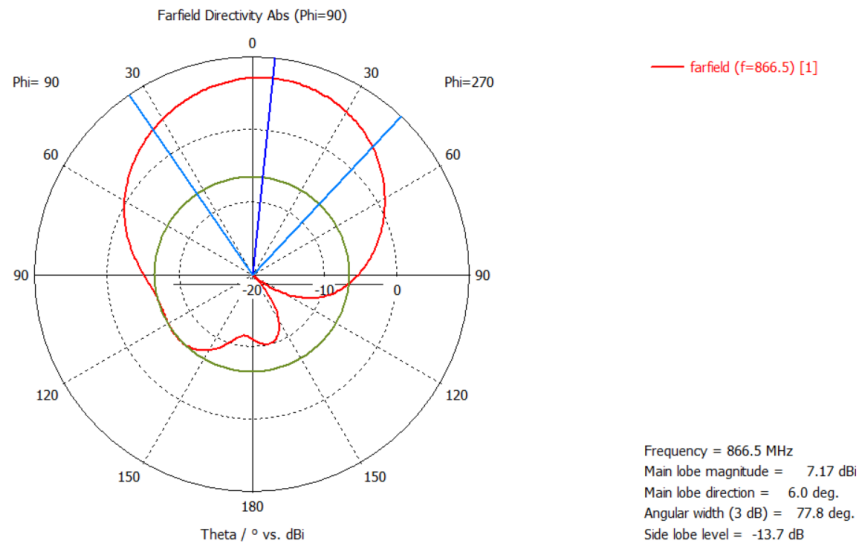


Figure 7 — 2D directivity cut, $\phi = 90^\circ$, at 866.5 MHz

The 2D cut confirms a clean single main lobe: main-lobe magnitude 7.17 dBi, main-lobe direction 6.0° off boresight, 3 dB angular beamwidth 77.8° , and side-lobe level -13.7 dB — the narrow, single-lobed pattern needed to avoid reading tags in an adjacent lane. The array pattern (Figure 6) shows the expected further narrowing and directivity increase from combining the two elements.

3.5 Structure, Surface Current and Layout

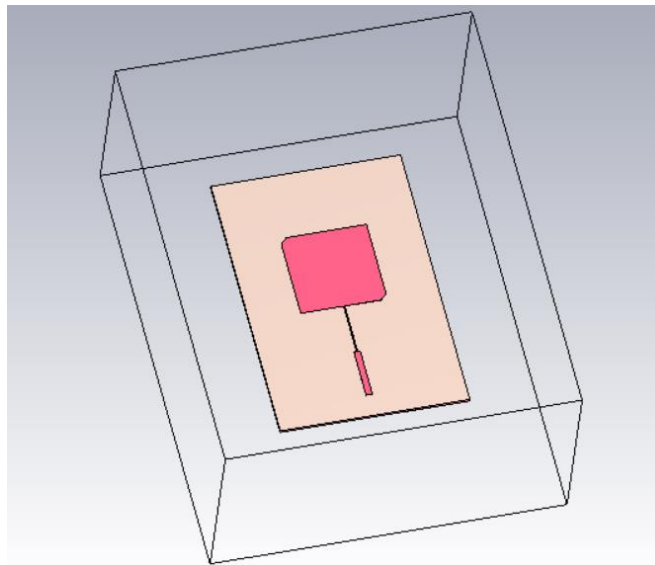


Figure 8 — 3D model: patch, feed line and substrate/ground stack

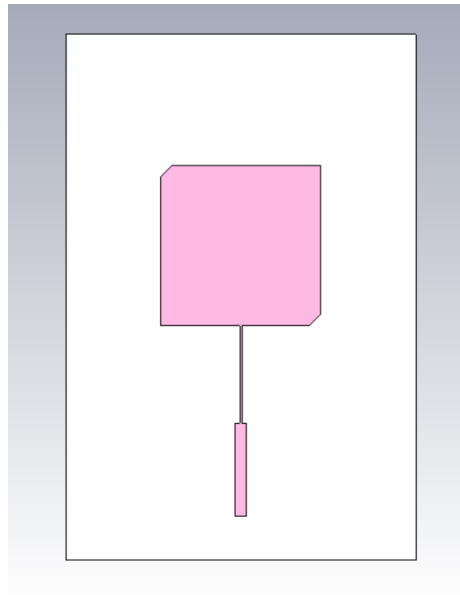


Figure 9 — Top view: corner-truncated patch with quarter-wave transformer feed

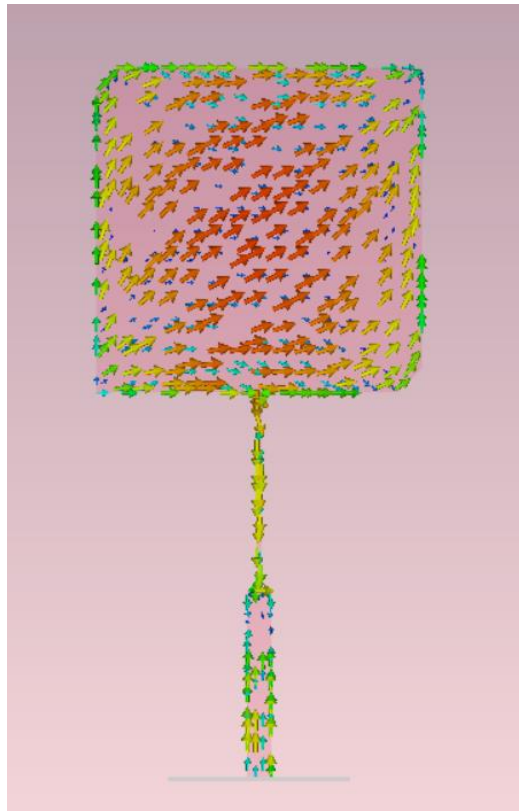
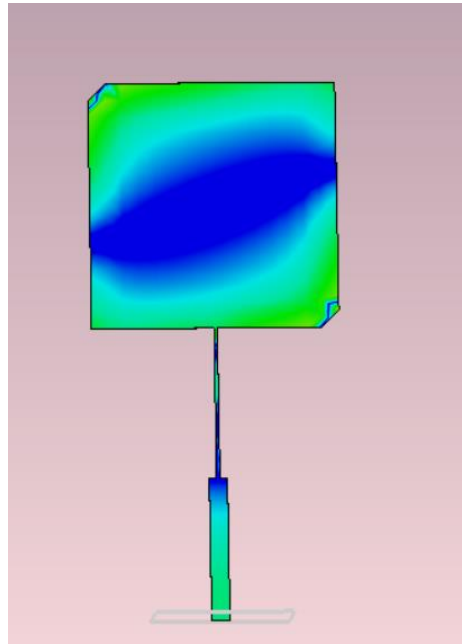


Figure 10 — Surface current distribution on the patch



Electric field distribution

The surface current plot shows current flowing diagonally across the patch with comparable magnitude along both edges near the truncated corners — the signature of two orthogonal, equal-amplitude degenerate modes in quadrature, which is the mechanism producing circular polarization from the single feed point.

4. Conclusion

The corner-truncated RO4003C patch, fed through an integrated microstrip quarter-wave transformer, meets the ETSI 865–868 MHz band requirement with S_{11} below -17.7 dB and VSWR below 1.3 across the band, and achieves circular polarization (axial ratio ≤ 3 dB across most of the band) from a single feed point. The single element alone provides 7.17 dBi, short of the 8–9 dBi target; the 4-element array raises this to 9.78 dBi while narrowing the beam and keeping side-lobe level at -13.7 dB, directly satisfying the narrow-beam / minimal-side-read requirement of the boom-barrier lane application. The large 300 mm $\times$ 200 mm ground plane additionally shields the patch from the metal pole mount and suppresses back-lobe radiation. The array configuration is therefore recommended as the final design for deployment.